

CatDB: Data-catalog-guided, LLM-based Generation of Data-centric ML Pipelines

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ABSTRACT

Data-centric machine learning (ML) pipelines extend traditional ML pipelines—of feature transformations, hyper-parameter tuning, and model training—by additional pre-processing steps for data cleaning, data augmentation, and feature engineering to create high-quality data with good coverage. Finding effective data-centric ML pipelines is still a labor- and compute-intensive process though. While AutoML tools use effective search strategies, they struggle to scale with large datasets. Large language models (LLMs) show promise for code generation but face challenges in generating data-centric ML pipelines due to private datasets not seen during training, complex pre-processing requirements, and the need for mitigating hallucinations. These demands exceed typical code generation as it requires actions tailored to the characteristics and requirements of a particular dataset. This paper introduces CatDB, a comprehensive, LLM-based system for generating effective, error-free, and efficient data-centric ML pipelines. CatDB leverages data catalog information and refined metadata to dynamically create dataset-specific rules (instructions) to guide the LLM. Moreover, CatDB includes a robust mechanism for automatic validation and error handling of the generated pipeline. Our experimental results show that CatDB reliably generates effective ML pipelines across diverse datasets, achieving accuracy comparable to or better than existing LLM-based systems, standalone AutoML tools, and combined workflows of data cleaning and AutoML tools, while delivering up to orders of magnitude faster performance on large datasets.

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PVLDB Artifact Availability:
The source code, data, and/or other artifacts have been made available at https://github.com/CoDS-GCS/CatDB.git.

1 INTRODUCTION

Modern data-driven applications increasingly rely on machine learning (ML) for cost-effective intelligence augmentation and intelligent infrastructure in various domains, such as health-care, finance, transportation, and production [39]. The traditional data science lifecycle is exploratory and includes formulating various hypotheses, integrating the relevant data, as well as developing and evaluating multiple predictive ML pipelines [16, 21].

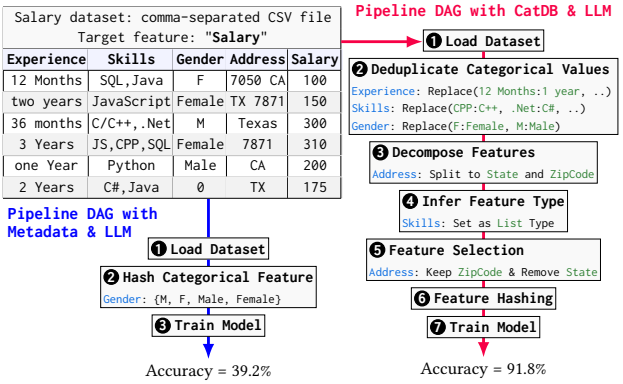


Figure 1: Comparison of prompt engineering for data-centric ML pipelines: [Metadata-only & LLM] versus [CatDB & LLM]. While the former struggles (39.2%), CatDB customizes instructions based on data characteristics to guide the LLM.

Data-centric ML Pipelines [57] extend traditional ML pipelines [1, 7] by additional pre-processing steps for data validation [68], data cleaning [30, 48, 74], data augmentation [22, 45], and feature engineering [63, 70]. Creating a high-quality dataset with good coverage often yields better accuracy than advanced ML models [53] or neural architecture search [60]. Even the original AlexNet paper [45] that won the ImageNet 2012 challenge and re-ignited research on deep neural networks (DNNs) heavily relied on data augmentation. Unfortunately, devising effective data-centric ML pipelines is a labor- and compute-intensive process. For example, tuning data augmentation [22] or data cleaning [74] pipelines requires expensive reinforcement learning or evolutionary algorithms.

Limitations of LLMs and AutoML Tools: LLMs, such as GPT-4 and Gemini-2, are increasingly utilized for tasks like code generation and data wrangling [55]. AutoML tools optimize feature selection and model tuning [24, 27, 46], but struggle to scale with large datasets due to time-intensive data analysis and the complexity of hyper-parameter search spaces. In contrast, LLMs generalize across diverse coding tasks and adapt ML pipelines dynamically. For well-known datasets, such as Titanic, LLMs like GPT-4 and Gemini-2 generate accurate and effective ML pipelines even without explicit dataset characteristics in the prompt. Readers can test this by prompting an LLM with: "Generate a data science pipeline for the Titanic dataset." Since Titanic is a widely used benchmark dataset that LLMs are typically exposed to during training, they can retrieve and apply relevant pre-processing steps, feature engineering, and model choices with high accuracy. However, LLMs struggle

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